GUC

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Internship Evaluation & Reporting

Thank you for taking the time to complete this form, this evaluation will be used to assess the student's participation in the internship program.

Supervisors, using the form below please evaluate the student who interned with your organization, institution, or business. You can fill out this form electronically or you can fill it manually but eventually it must be signed and stamped from the company's side.

Please note that part I & III should be completed by the intern, part II should be completed by the direct supervisor in the company.

Part I. GENERAL INFORMATION – STUDENT'S INPUT

Student Info:

Student Name: Ahmed Amir Mansy Mohamed__ GUC Student ID No.:55-17668________

Faculty: IET________ Major: Networks Engineering________

Student Mobile No.:01112022157________

Internship Info:

Company Name: SSTM-Simply Smart IOT Solutions________

Core Industry/Business: Smart IOT Solutions Country: Egypt________

Supervisor Name: Amr Talaat______ Supervisor Job Title: Founder & Managing Director_____

Supervisor Tel. No.: ________ Supervisor Mobile No.: +201001400995________

Supervisor E-mail: amrtalaat@sstm-eg.com Training Department(s): IOT Software Engineering________

Source of internships: (1) SCAD office (2) on my own (X) Referrals from GUC TA/Dr. (4) Recruitment website (5) others:

Work Place: (X) Organization (2) Head Office (3) Branch (4) Factory (5) Site (6) Others:

Part II. EVALUATION AND COMMENTS – DIRECT SUPERVISOR'S INPUT

Period of Internship (dd/mm/yyyy) (dd/mm/yyyy)

From: 15__/06__/2024_____ To: 07__/09__/2024____

Internship nature (Enrollment Status)

☐ Part time Please specify, no. of Days per week:________ hours per day :________

X Full time Please specify, no. of Days per week:5________ hours per day :8________

Company Stamp

S.S.T.M

For SCAD internal use only

Serial no.	SCAD Comment	Academic Reviewer Comment	Academic Reviewer Signature
			☐ Accepted ☐ Rejected Reason of rejection: Signature:

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Please evaluate student's performance by marking the appropriate box:

For each of the following aspects, please mark the box in the rating scale that most closely corresponds to your evaluation of the profile of the student during the internship period. Please also feel free to offer comments and suggestions for changes and improvements in the space provided at the end of the form.

1=Unsatisfactory	2=Below Average	3=Satisfactory	4=Above Average	5=Excellent

	1	2	3	4	5	NA
Skills & Professional Attributes						
Ability to adapt to change				X		
Analytical skills					X	
Collecting data/ research data skills					X	
Creativity				X		
Follow up skills					X	
Interpersonal skills with peers, supervisors, and clients					X	
Problem solving				X		
Punctuality				X		
Reporting skills					X	
Responsibility and accountability				X		
Stress handling				X		
Taking initiatives				X		
Teamwork					X	
Time management				X		
Other:						
Technical Background						
Technical Knowledge				X		
Compatibility of technical skills with the job				X		
Other:						
Command of Languages						
Arabic					X	
English				X		
German						
Other:						

1=Unsatisfactory	2=Below Average	3=Satisfactory	4=Above Average	5=Excellent

	1	2	3	4	5	NA
Computer Programs & Databases **Please use space below in specifying the program/software used during the internship and evaluate student's performance accordingly**						
LoRa WAN Implementation					X	
C++					X	
Network Optimization					X	

Overall Evaluation of Student's performance and profile					
Unsatisfactory	Improvement needed	Meets expectations	Exceeds expectations	Exceptional	NA
		X			

General Comments & Recommendations: (kindly mention intern potentials, areas of further development or technical constraints encountered during the internship period)

- study more approaches for data engineering other than hidden maKoV model.
- study market eperience for data engineering.

	Yes	No	Maybe
Do you think similar candidates would fit in the Organization culture and qualify for job needs?	X	☐	☐

Student Signature:

Ahmed.Amir.Mansy______________

Date:

15/06/2024→07/09/2024

Supervisor Signature:

Amr Talaat

Date:

15/06/2024→07/09/2024

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Part III. INTERNSHIP REPORT – STUDENT'S INPUT

- This report has to be prepared by the student, it must be prepared and written in a <u>computerized</u> format, submitting the report in hand written format will not be accepted.
- Kindly refer to the Internship Report writing Guidelines on the GUC intranet – SCAD office folder.
- This report will be reviewed and evaluated from internal faculty members.

Internship Title: SSTM LoRa WAN Internship(IOT Data Collection and Cloud Integration Internship)

Company / Organization Name: SSTM-Simply Smart

Introduction: (Not less than 100 words) (should depict the main purpose of the report, covering the objective out of performing this internship in this industry/company specifically then cover the outline for the report's structure)

Company / Organization Description: (Not less than 100 words)

Internship Performed Tasks: (Not less than 100 words)

Internship Evaluation: (Not less than 100 words) (This section should answer the following questions in the form of a paragraph)

What skills do you think that you have gained from the internship? Did the internship meet your expectations? If not, please explain why? How do you think the internship will influence your future career plans? How do you think the internship activities that you carried out are correlated with your studies? Which of the academic courses that you have taken in GUC were the most related to your internship?

Conclusion: (Not less than 100 words) (A summary of key conclusions derived from the internship experience. general observations about the sector in which your internship company/institution operates)

Please rate your satisfaction with the internship experience.

X Very satisfied ☐ Somehow satisfied ☐ Neutral ☐ Somehow dissatisfied ☐ Very dissatisfied

Would you recommend this internship to other colleagues?

X Yes ☐ No ☐ Maybe

References: (If any external sources are used, provide references for any information quoted)

Appendices: (Upon availability, charts, pictures, etc.)

Disclosure / Confidentiality Agreement

This agreement is to acknowledge that the information provided by any company / organization during the internship is unique to this business and confidential.
Therefore, anyone reading this agreement agrees not to disclose any of the information provided during the internship without notifying & taking the employer's / supervisor's approval.

Internship Title:

SSTM LoRa Internship (IoT Data Collection and Cloud Integration Internship).
Company / Organization Name:

SSTM simply smart.

Introduction:
SSTM LoRa Internship is a hands-on, project-based program designed for us Networks and Communication engineers. Over the course of the internship period, we will dive into the world of the Internet of Things (IoT) by focusing on two cutting-edge technologies: Bluetooth Low Energy (BLE) and LoRa (Long Range) communication. Participants will research and implement solutions for scanning BLE beacons using ESP32-C microcontrollers and transmitting the data packets to the cloud using the LoRaWAN protocol. By the end of the program, interns will have developed a fully functional system capable of monitoring data from the beacon and sending it using the LoRaWAN protocol.

Company / Organization Description:
SSTM Egypt was established with the objective of developing IoT industry-grade solutions. SSTM's founders have vast experience in research, electronics design, information technology, and product development. Over the years, SSTM has undertaken numerous R&D initiatives in partnership with multinational organizations, achieving high-quality results within budget, operational target prices, and detailed specifications. SSTM has the advantage of fully understanding the ecosystem and every sub-component, making it the ideal provider of machine-to-machine (M2M) and Internet of Things (IoT) platforms. With excellent support capabilities, SSTM can develop custom solutions across every element of the value chain, from GSM to Zigbee, Bluetooth 5.0, and even wired components as needed.

Internship Performed Tasks:
Our internship began with research and setup. We explored BLE technology, delving into its fundamentals and understanding how BLE beacons function. I identified different types of BLE beacons and their practical applications, while also studying LoRa technology to grasp its core principles and compare it with other wireless communication methods. We reviewed the ESP32 and ESP-C, learning about their architecture, capabilities, and programming environments. To prepare for development, we installed and configured the necessary software tools and libraries, and set up and tested our hardware components. Next, we shifted our focus to BLE scanning implementation. I wrote code to detect nearby BLE beacons using the ESP32 and explored various scanning methods. This phase also involved data parsing and extraction, where I developed code to process and extract

relevant data from BLE beacon signals while documenting different data structures and formats. Testing and debugging were critical to ensure that our BLE scanning code performed reliably across different environments and to resolve any detection or data parsing issues. Finally, we tackled LoRa communication implementation. I developed code to transmit data over LoRa from the ESP32, selecting appropriate LoRa libraries and protocols. We then integrated BLE scanning with LoRa transmission, ensuring a smooth data flow from beacon detection to cloud communication. The entire system underwent rigorous testing and optimization to assess its performance, reliability, power efficiency, and transmission speed.

Internship Evaluation:

I gained multiple skills, such as implementing a network protocol using Visual Studio Code, as well as working with long-range communication technology using a new programming language for me—C++. Additionally, I learned about code migration between different languages, as the Bluetooth libraries only worked with Python, requiring us to link components between the two languages. On the other hand, we also had to develop brand-new libraries in Python. Working with Bluetooth Low Energy (BLE) beacons was truly fascinating, and using a microcontroller for the first time was an exciting experience. The internship met my expectations by helping me develop practical skills in a team-oriented environment. The senior engineers we worked with shared their extensive knowledge and expertise, making the experience even more enriching. To be honest, the internship made me reconsider my perspective on working with hardware. I always thought hardware was difficult and unforgiving—one mistake, and everything could be lost. However, this was not the case at all. Moreover, I realized that working with the right team and maintaining effective communication leads to the fastest and best results. As a Networks Engineering student, I highly value the knowledge I gained, as it aligns well with what I studied. The work on network protocols, IoT systems management, and wireless network traffic management closely relates to network protocols, random signals, noise, and networking labs. It also connects with probability-based courses such as modeling and simulation and performance modeling.

Conclusion:

The SSTM LoRa Internship provided an invaluable, hands-on experience in the dynamic field of IoT. Throughout the program, I gained practical skills in BLE scanning, LoRa communication, and network protocol development. Working with both Python and C++ to manage code migration challenges deepened my understanding of integrating diverse technologies. I learned that hardware engineering, once intimidating, is a field ripe with innovation when approached with teamwork and effective communication. The guidance from experienced engineers was pivotal, offering insights into real-world debugging, project management, and best practices in system optimization. Observing SSTM's expertise in deploying scalable, interconnected systems highlighted the need for adaptability and customization in today's IoT landscape. This internship not only reinforced my academic foundation in Networks Engineering but also reshaped my perspective on hardware, revealing it as an exciting and accessible domain. Overall, this experience has prepared me to contribute confidently to the IoT sector, underscoring that collaborative innovation and continuous learning are key to overcoming technical challenges and achieving success in rapidly evolving technological environments.

References:

c++ faq - The Definitive C++ Book Guide and List - Stack Overflow, C++ Programming Language - GeeksforGeeks, https://www.mdpi.com/1424-8220/21/23/7992, https://www.thethingsnetwork.org/docs/lorawan/what-is-lorawan/, What is LoRaWaN | PPT, LoRaWAN® Specification v1.0.3, LoRaWAN® Specification v1.1, For Seeed nRF52 Boards Library | Seeed Studio Wiki, Getting Started with Seeed Studio XIAO nRF52840 Series | Seeed Studio Wiki.

Appendices:

Cannot be disclosed due to confidentiality and disclosure agreements.